Information Structure Motivates the Use and Emergence of Non-canonical Constructions

Moshe Poliak moshepol@mit.edu, Giuseppe Ricciardi, Edward Gibson

Introduction. A truism in linguistics is that no true synonyms exist[1]. And yet, language has pairs of constructions with very similar meanings[†]. Why would natural language avoid true synonyms, which may be inefficient, and yet allow for multiple alternations to exist? Moreover, one construction in an alternation is always more canonical than the other, so why use non-canonical constructions at all? We propose an explanation rooted in information structure. An old observation is that languages tend to position old (given) information before new information[2-10]. Furthermore, it has been proposed that, in pairs of constructions that have similar meanings but different word orders (alternations), the non-canonical construction is associated with a constrained information structure[11-12]. In English, a well-known example is the dative alternation, where the prepositional object construction shows no preference for either information structure **(Ex. 1-a-i=1-b-i)**, whereas the double object construction shows a strong preference for old-before-new (**Ex. 1-b-ii>1-a-ii**)[13-14]. Following the insights of Halliday, Chafe, Givón, Birner, and Ward, we propose that **non-canonical constructions are used to achieve old-before-new information structure by providing an alternative word order, as long as the non-canonical construction is frequent enough to be accessible.**

Method. We conducted 4 online experiments, one per alternation and language: dative and locative in English (MTurk Ns=63,55) and in Italian (Prolific Ns=50,50). On each trial, we presented participants with a context sentence followed by a target sentence that was either canonical or not (**Ex. 1-3**). The context sentence mentioned one, two, or none of the arguments in the target sentence. Old arguments in the target sentence were definite, and new arguments were indefinite. Thus, each experiment had a 2x2x2 design: the information status of the 2 arguments and the canonicity of the constructions. Participants were asked to rate how natural the target sentence sounds on a scale of 1 to 5. We also evaluated the frequency of the non-canonical construction using a corpus analysis of COCA[16], automatically parsing dependency structures using Stanza[15].

Results & Discussion. For brevity, we consider only sentences with new-old and old-new information structure (**Figure 1**). For each experiment, we fit a Bayesian ordinal regression with the full random effects structure and the default priors from brms. We tested for a negative interaction between canonicity and information order, which indicates that the old-before-new preference is greater in the non-canonical construction. This interaction was detected within the dative alternation (95% CrI= [-2.21, -.67]) but not in the locative alternation (95% CrI = [-1.04, .40]). However, the corpus analysis showed that the locative inversion **(Ex. 2-b)** is rare (1.4% of all locatives), and thus is potentially not a cognitively available construction in the alternation. In contrast, in Italian, the non-canonical constructions in the dative and locative alternations are relatively frequent (25.6% and 33.9% of all datives and locatives, respectively). Thus, we repeated the experiment using the same design, but with the materials translated to Italian. We found a substantial negative interaction for both the dative (95% CrI=[-2.51, -.83]) and the locative (95% CrI=[-2.89, -.90]). These results suggest an answer for why it is the case that 2 frequent constructions co-exist in language: the non-canonical construction is used as a way to position old information before new information in the sentence.

[†]The terms *alternation* have been used in the past to suggest that one construction is derived from the other or that they have identical meanings, a notion that we do not argue for[CITE]. By calling a pair of constructions an alternation, we simply mean that these two constructions have similar enough meanings such that they can be used interchangeably in most contexts.

Examples

1. English Dative
    a. Context (introduces theme): An understudy for a new Broadway show began conversing with a violinist.
        i. PO (canonical): The understudy showed a notebook to the violinist.
        ii. DO (non-canonical): The understudy showed the violinist a notebook.
    b. Context (introduces recipient): An understudy for a new Broadway show kept a notebook to document the show's progress.
        i. PO (canonical): The understudy showed the notebook to a violinist.
        ii. DO (non-canonical): The understudy showed a violinist the notebook.
2. Locative alternation:
    a. Locative NP-V-Loc (canonical): A weapon lay behind a box.
    b. Locative Loc-V-NP (non-canonical) Behind a box lay a weapon.
3. Italian constructions:
    a. Dative NPPP (canonical): La configuratora mostrò un quaderno al violinista
    b. Dative PPNP (non-canonical): La configuratora mostrò al violinista un quaderno
    c. Locative NP-V-Loc (canonical): Un'arma giaceva dietro a una scatola
    d. Locative Loc-V-NP (non-canonical): Dietro a una scatola giaceva un'arma

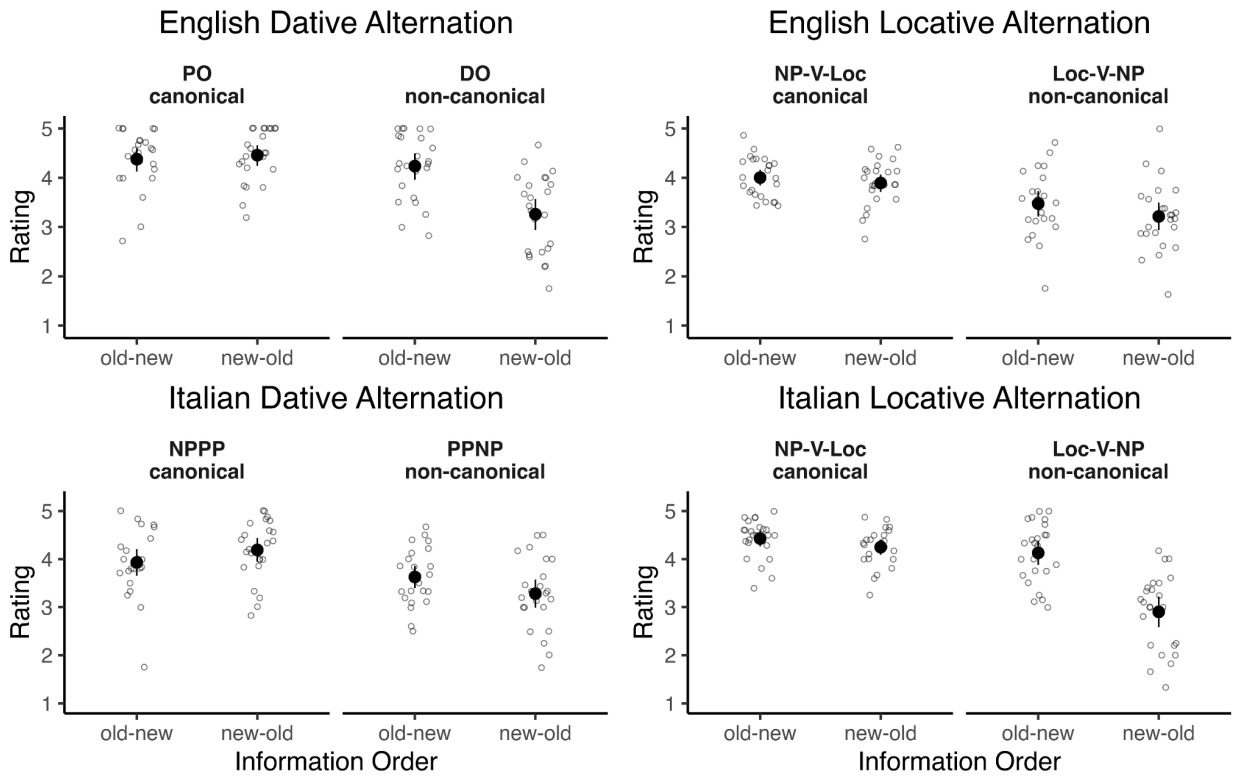

Figure 1. The results of the 4 experiments. Hollow circles are item means, and error bars are 95% bootstrapped confidence intervals over item means.

References: [1]Haiman, 1980; [2]Halliday, 1967; [3]Firbas, 1966; [4]Chafe, 1976; [5]Clark & Clark, 1977; [6]Clark & Haviland, 1977; [7]Prince, 1981; [8]Gundel, 1988; [9]Vallduví, 1990; [10]Arnold et al., 2000; [11]Givón, 1983; [12]Birner & Ward, 2009; [13]Clifton & Frazier, 2004; [14]Brown et al., 2012; [15]Qi et al. 2020; [16]Davies, 2008